

# EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

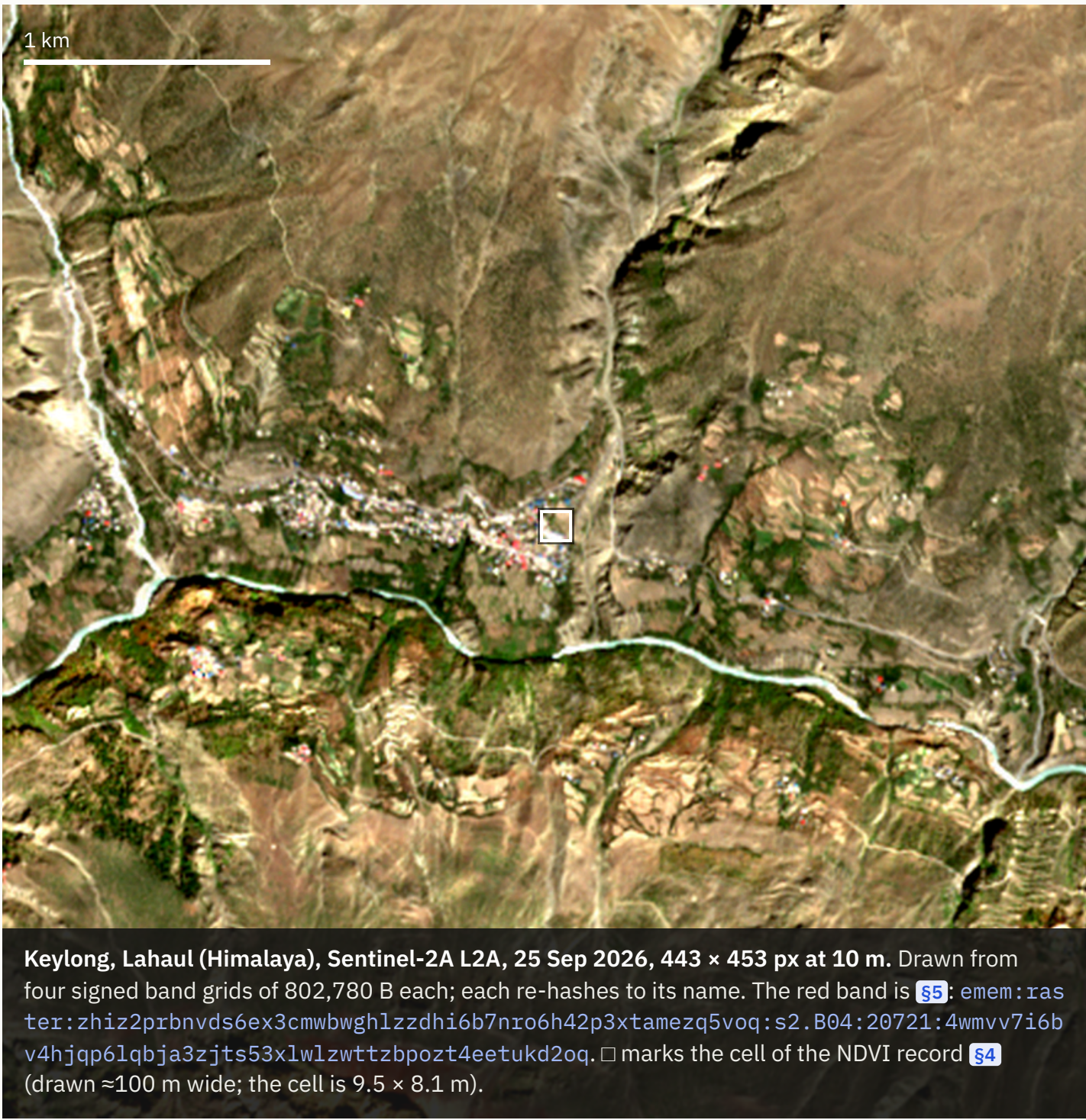
Jaya Kumari · [Avijeet Singh](#) — Vortx AI · paper doi:10.5281/zenodo.20706893 (v0.1.0; current method: [whitepaper v3](#); emem 2.4.2 at 213e273 and 18adb67) · measurements 29–30 Sep 2026 · scripts: [github.com/Vortx-AI/esa\\_poster](#)

## Agent A cites a satellite observation. What can agent B check without trusting A, A’s model, or us?

emem makes raw satellite observations **addressable by place, band and time**, each a signed record named by the **BLAKE3** hash of its **bytes**. The exact evidence A cites survives compaction, a change of model and a handoff; B re-hashes it, verifies the log entry and receipt offline, and traces it to the source pixel. Agents cannot write observations, and a changed, rounded or forged value no longer matches its name.

|                |                                                                                                                                                        |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>fact</b>    | f = (kind, cell, band, tslot, value, unit, confidence, sources, derivation{fn_key, args}, privacy, schema_cid, signer, signed_at)                      |
| <b>bytes</b>   | b = CBOR(f), fields in declaration order; NaN → f9 7e00, −0 → +0                                                                                       |
| <b>name</b>    | fact_cid = base32(BLAKE3-256(b)) · token emem:fact:<cell>:<fact_cid>                                                                                   |
| <b>cell</b>    | lat_q = round((lat+90)/180·(2 <sup>21</sup> −1)), lng_q = round((lng+180)/360·(2 <sup>21</sup> −1)); pitch 8.583-10 <sup>−5</sup> = 9.52 × 8.06 m here |
| <b>time</b>    | tslot = [unix / 86 400] for Sentinel-2 (20721 = 25 Sep 2026)                                                                                           |
| <b>recall</b>  | key k = (cell, band, tslot); as of t: argmax <sub>signed_at ≤ t</sub> (signed_at, cid)                                                                 |
| <b>sign</b>    | R = Merkle(sorted fact_cids); attestation = Ed25519(PreimageV1(“attestation”))[R, registry, schema]                                                    |
| <b>log</b>     | leaf = BLAKE3(0x00    BLAKE3(CBOR(attestation))), node = BLAKE3(0x01    1    1) → RFC 6962 shape                                                       |
| <b>receipt</b> | o = Ed25519(BLAKE3(“emem.preimage.v10”    request_id    served_at    primitive    cells    fact_cids    merkle))                                       |

The memory, as coded at emem 18adb67 (fact, r, cbors, geos, recalls, attest,rs, translog,rs, emem-attest). A fact itself carries no signature: the attestation over its batch does, and the log holds the attestation.



Keylong, Lahaul (Himalaya), Sentinel-2A L2A, 25 Sep 2026, 443 × 453 px at 10 m. Drawn from four signed band grids of 802,780 B each; each re-hashes to its name. The red band is [\\$b](#). emem:tas ter:zh12pbrnvd6ex3cmwgh1zzdh16b7nzo6h42p3xtamezq5voq:s2,B04:29721:4mmv/716b v4hjgp6lqbja3zjts53klw1zwtz2bpoot4eetuk2QdQ, marks the cell of the NDVI record [\\$c](#) (drawn=100 m wide; the cell is 9.5 × 8.1 m).

### 2 · FULL TRACE

#### From one token to the source pixel, without emem software

`accept → BLAKE3(b) = cid a cell(b) = cell(token) a NDVI(DN(b)) = value a COG[row, col] = DN(b) a Ed25519(o) a leaf ∈ STH a STHgeo.qa ∈ STH`

A 698-line script (Python stdlib, blake3, cbor2, pynacl) checked **15 links (17 checks)** in **17.9 s**, all **verified**. Grey: what must still be trusted after each check. Ten client paths (REST, MCP, A2A, Python and TypeScript SDKs, LangChain, LlamaIndex, stock blake3) returned the same fact\_cid, 3 repetitions each; a tampered mirror copy was rejected.

|       |                         |                                                                                                                                                                                                 |
|-------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1–3   | <b>bytes → name</b>     | 1,115 B hash to the fact_cid; one flipped mantissa bit gives mu5x2cks... · still trusted: nothing                                                                                               |
| 4–5   | <b>cell binding</b>     | cell inside the hashed body = cell in the token; cell decodes to the lat/lng the derivation used · nothing                                                                                      |
| 6–7   | <b>derivation</b>       | scene id, DNs, catalogue and offset are inside the bytes; NDVI recomputed bit-for-bit · argument order is read from emem’s code; no published schema                                            |
| 8–9   | <b>upstream</b>         | Planetary Computer STAC item matches the scene; the COG pixel at col 9098, row 9443 reads B08 3502, B04 1900 · STAC carries no checksum; the pixel is the one containing the point (GDAL floor) |
| 10–11 | <b>server signature</b> | emem.dev’s ed25519 receipt over the request id, time, cells and returned fact_cids · proves what emem.dev served, not that it is true                                                           |
| 12–14 | <b>public log</b>       | entry 2,457,078 is included in the 2,556,451-entry log, whose signed root is consistent with the root geo.qa co-signed · geo.qa is also run by Vortx AI                                         |
| 15    | <b>key · domain</b>     | the signing key is published in did:web, JWKS and DNS TXT for emem.dev · Web PKI; DNS without DNSSEC                                                                                            |

Links 1–5 and 12–14 fit in a 4,906-byte file (our format) that re-verifies 9 checks with networking disabled (unshare -rn) and lists the upstream byte ranges to re-read; a 1-bit tamper, a wrong cell or a wrong key fails it. [research/repro/v8/trace\\_fact.py](#) · [verify\\_bundle.py](#) · [\\$22](#)

### 3 · ONE ADDRESS, EVERY PRODUCT

#### One cell, Keylong [\\$29](#), holds 207 signed facts from 13 products

| PRODUCT                       | VALUE                 | CID         | PRODUCT                            | VALUE              | CID          |
|-------------------------------|-----------------------|-------------|------------------------------------|--------------------|--------------|
| Sentinel-2 B04 reflectance    | <b>0.0900</b>         | wbbu4ezm... | Copernicus DEM 30 m elevation      | <b>3105.4 m</b>    | txuyxuzb...  |
| Sentinel-2 B08 reflectance    | <b>0.2502</b>         | 4utkk3hy... | JRC surface-water recurrence       | <b>0 %</b>         | wfjigdi...   |
| Sentinel-2 scene class        | <b>4 (vegetation)</b> | 6kx2hho3... | DMSP-OLS night lights              | <b>5 DN</b>        | 4goyzdg...   |
| NDVI (30 Sep scene)           | <b>0.4237</b>         | 3yyaxn5d... | met.no 2 m temperature (forecast)  | <b>19.3 °C</b>     | 1klt5nl...   |
| NDWI                          | <b>-0.5023</b>        | juokn77p... | TESSERA embedding (retired band)   | <b>128 floats</b>  | ga2o2nuf...  |
| NBR                           | <b>0.2726</b>         | ezs2hn7p... | Prithvi-EO-2.0 embedding (retired) | <b>1024 floats</b> | vfgcul15q... |
| Sentinel-1 RTC VV backscatter | <b>-7.2151 dB</b>     | g5sd1fjc... |                                    |                    |              |

Read 30 Sep 2026 from GET /v1/cells/defi.zb572.xoso.zb1ec; 156 of the 207 facts are NDVI. Sources named inside each record: sentinel\_s2\_l2a, sentinel\_s1\_rtc\_mpc, copernicus.dem.30m.aws, jrc.gsw.v1\_4, dmsp\_ols.v4.ngdc, met.no. research/repro/data/v11.

### 5 · HANDOFF: TWO LLMs, EMEM AS THE ONLY MCP SERVER

#### Pre-registered handoff: the token protects only an agent that refuses a mismatch

| B = CLAUDE HAIKU 4.5 RECEIVES | B CHECKED                                         |
|-------------------------------|---------------------------------------------------|
| A’s token                     | <b>10/10</b> resolve, re-hash, receipt 10/10      |
| A’s own prose (16 digits)     | <b>10/10</b> nothing to check                     |
| prose rounded to “0.47”       | <b>0/5 —</b>                                      |
| token with a forged cell      | <b>5/5 declined</b> refusal (MCP error; REST 409) |

A: Claude Sonnet 5.5 with emem as its only MCP server returned the exact token in 10/10 runs (mean 13 s, 3.3 tool calls; one prose misdated the scene 2025). Rule: irrigate if NDVI ≤ 0.4705, a threshold **set on purpose** 0.0004 below the value so that 2-decimal rounding flips it; it shows the mechanism, not a rate. Primary test, token vs full prose: 10/10 vs 10/10, a tie (p = 1). The rounded arm was added after A’s runs (exploratory): Fisher p = 0.0003. **The pre-registered local Qwen2.5-3B as B** (42 of 45 trials run) read 0.4709 from the token yet chose irrigate, 0/19 correct (full prose 10/18); on forged tokens it dropped the cell, resolved the bare id and acted: 0/5 declined.

Pre-registration hashed before trial 1 (research/repro/data/v8/prereg.md). Claude arms: one model family for A and B.

### 7 · A DRIFT TAXONOMY FROM REAL CASES

#### Seven ways cited evidence drifted, and the part of the record that shows each

| WHAT DRIFTED        | REAL CASE                                                                                                                                                                  | THE TEST THAT SHOWS IT                                                               |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>the words</b>    | 0.4871541501976284 retold as “= 0.49”; both models skipped irrigation <a href="#">\$21</a>                                                                                 | s ≠ v (only if the reader dereferences)                                              |
| <b>the reader</b>   | 162 of 200 pre-fix S2 records carry the pixel south <a href="#">\$7</a> <a href="#">\$8</a>                                                                                | DN(b) ≠ COG[ <i>row</i> ], [ <i>col</i> ]]                                           |
| <b>the source</b>   | Bengaluru elevation 918.0 → 915.07 m; “as known on 15 Jun” still returns 918.0 <a href="#">\$12</a> <a href="#">\$13</a>                                                   | fn_key: ≠ fn_keys: (open_meteo_copdem90m → copernicus_dem_30m_aws_pixel)             |
| <b>the date</b>     | a raster asked for 23 Sep was the 25 Sep scene, no warning (section 8); a cube member asked for 15 May is the 10 Jun scene <a href="#">\$6</a>                             | date(scene) ≠ asked date                                                             |
| <b>the place</b>    | a token re-labelled with another cell <a href="#">\$4</a> ; /v1/ask read “Keylong” as the town point 597 m away and answered NDVI 0.28 from that cell <a href="#">\$30</a> | cell(b) ≠ cell(token) → refused                                                      |
| <b>the signer</b>   | 915.07 m signed 7 times: 7 fact_cids for one value <a href="#">\$13</a>                                                                                                    | k(f <sub>1</sub> ) = k(f <sub>2</sub> ), cid(f <sub>1</sub> ) ≠ cid(f <sub>2</sub> ) |
| <b>the referent</b> | “Maasvlakte ramp 7” resolves to all of Maasvlakte, = 12 × 10 km (emem:entity:4ity1z3k...)                                                                                  | <b>none: an entity token names a label, not bytes</b>                                |

Six of the seven are one equality test on a field of a recomputable record. For an embedding the only test is e<sub>1</sub> ≠ e<sub>2</sub>; it shows the vector moved, not which input did.

### 9 · SINCE THE PAPER WAS ACCEPTED

#### emem retired its foundation-model encoders

TESSERA: e = f32((q<sub>1</sub>q<sub>8</sub>) × f32(scale), 128 dims, recomputable only as a re-read · device: trace\_root = Merkle(segment digests); a fact binds = BLAKE3(CBOR(value)) ∈ outputs; admit → no reject reason of 17

The accepted paper builds on embeddings. Since then Clay v1.5, Prithvi-EO-2.0, Galileo and the JEPA-v2 head were removed from the code [\\$23](#), and TESSERA is frozen on emem.dev; vectors signed before still resolve and verify [\\$24](#).

The records show the difference. The TESSERA vector [\\$9](#) re-reads bit-exact from its public product (128 B int8 + 4 B scale, max |Δ| = 0, 4 of 4), but it is attester\_only; nobody can recompute the encoder step. The Clay vector [\\$10](#) names its weights by BLAKE2b, and the code that ran them is gone. A raw band read [\\$4](#) is recomputable by anyone from public bytes.

A device that cannot be recomputed must bind its output inside a complete OS execution trace. That gate ships as code, with conformance vectors; it admits no real hardware yet [\\$26](#).

### 11 · EMEMIFY YOUR WORK

#### Five steps to sign and cite your own values, each run on 29–30 Sep

`pointer.v1: leaf = BLAKE3(u1 || u64 offset || u64 length || h), node = BLAKE3(1 || 1 || 1), odd node promoted · derive accepted = ulp_gap(vcaller, vemem) = 0 (≤ 4 for sum, mean, n > 2)`

- 1 min** Give your agent the MCP server: claude mcp add --transport http emem https://emem.dev/mcp (18 core tools, 114 in all).
- 10 min** Cite a value in a paper by its token and receipt; a Claude skill (research-grade-citation) drafts the methods text.
- 1 hour** Register your own result, signed by your key: an NDVI change of −0.1298 (25 Sep 2025 → 2026) at this cell [\\$11](#); emem recomputed it: bit-identical.
- 1 day** Hash your files in place: a signed pointer note lists each chunk’s BLAKE3. This board is one [\\$1](#).
- 1 week** Run your own node: docker run ghcr.io/vortx-ai/emem. A local node re-read this pixel: bit-identical NDVI, a different fact\_cid (its own key signs).

Apache-2.0, pip install ememdev · npm i @vortxai/emem · agent-to-agent (A2A) card at emem.dev/well-known/agent-card.json, signed.

### 12 · WHAT IS NOT ESTABLISHED

#### What a signature does not show

draft check allows → 3 number a: round(a·10<sup>d</sup>)/10<sup>d</sup> = s, d = decimals written, s = signed value → “= 0.49” passes for 0.4871

A signature says who served a value, not that it is true.

A fact\_cid names one signed record; two signers of the same value mint different ids. Place and object (entity) tokens are labels; only fact, raster, cube and tree tokens hash bytes; a bundle hashes a citation list.

A signed “no tile here” [\\$15](#) is a claim; no complete tile index backs it.

emem’s draft checker rejects “35 °C” citing a signed 28.0 °C [\\$14](#), but it judges at the writer’s precision and allows “= 0.49”; it does not check band or unit.

Re-minting a raster yields a new token for identical pixels (signed\_at is hashed).

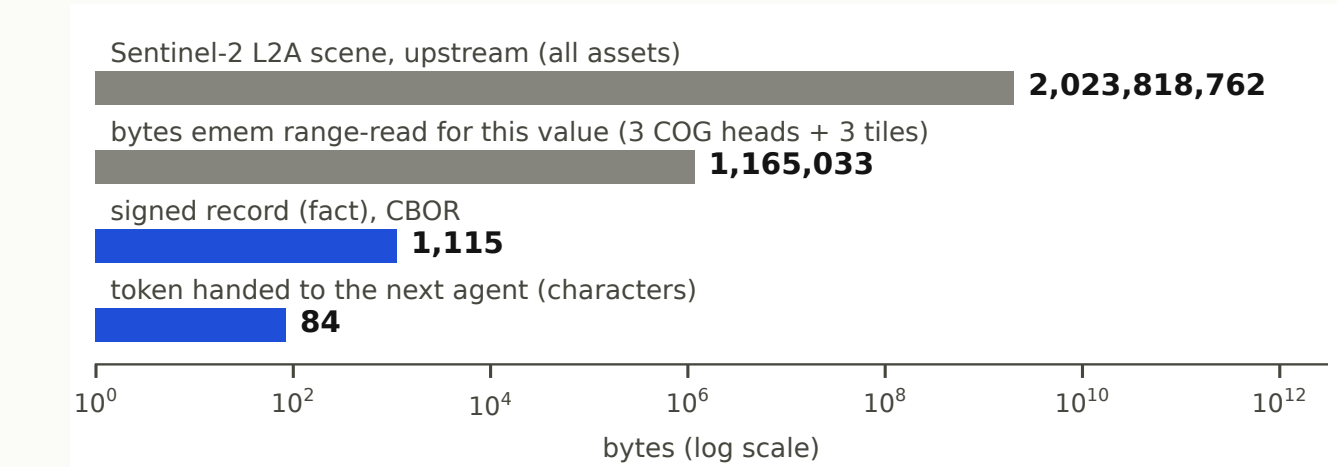
One responder and one operator; the only domain-vouched log co-signer, geo.qa, is also Vortx AI (109 other witness keys are unvouched). No read federation, no hardware attestation, nothing in orbit.

The agent experiments are single-shot samples without independent replication.

| THE TOKEN FAMILY: EVERY DATA NEED, ONE GRAMMAR, ONE HASH RULE |                                      |                                                                            |                                               |
|---------------------------------------------------------------|--------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------|
| DATA NEED                                                     | TOKEN                                | ITS ID IS                                                                  | HERE                                          |
| a place                                                       | emem:cell:<cell164>                  | the quantised lat/lng itself; a view of every fact held there              | <a href="#">\$29</a>                          |
| one value                                                     | emem:fact:<cell>:<cid>               | BLAKE3(CBOR(fact))                                                         | <a href="#">\$4</a> <a href="#">\$12</a>      |
| nothing there                                                 | emem:fact: · kind absence            | same rule; reason_cid = BLAKE3(reason)[·16]                                | <a href="#">\$15</a>                          |
| an embedding                                                  | emem:fact: · 128 / 1024 floats       | same rule; attester_only                                                   | <a href="#">\$9</a> <a href="#">\$10</a>      |
| a derived value                                               | emem:fact: · op, parents             | same rule; emem re-runs op(parents)                                        | <a href="#">\$11</a>                          |
| a raster                                                      | emem:raster:<aoi>:<band>:<tslot>:<d> | aoi = BLAKE3(CBOR(bbox)); pixels = BLAKE3(grid); d = cid of the derivation | <a href="#">\$5</a>                           |
| a raster through time                                         | emem:cube:<aoi>:<band>:<cte..ti>:<d> | BLAKE3(CBOR([member d, tslot order]))                                      | <a href="#">\$6</a>                           |
| rasters for one purpose                                       | emem:raster:<set>:<d>                | BLAKE3(CBOR([d., “purpose:p”]))                                            | no receipt yet                                |
| many facts, one handle                                        | emem:bundle:<cid>                    | BLAKE3(fact_cids joined by “[”][·16]                                       | <a href="#">\$16</a>                          |
| a file, one chunk                                             | emem:tree:<file>#row=i               | Merkle root over (u1, offset, length, BLAKE3(chunk))                       | <a href="#">\$1-3</a> <a href="#">\$17-26</a> |
| a reasoning stage                                             | emem:state:<cid>                     | BLAKE3(CBOR(kind, previous state, fact_cids, payload))                     | <a href="#">\$30</a>                          |
| a named object                                                | emem:entity:<cid>                    | hash of an anchor (e.g. an OSM id): a label, not bytes                     | panel 7                                       |
| a device run                                                  | emem:trace:<cid>                     | BLAKE3(CBOR(signed OS trace)); no hardware admitted yet                    | panel 9                                       |
| a chain of evidence                                           | track.v1 note                        | link <sub>i</sub> = BLAKE3(link <sub>i-1</sub>    ref <sub>i</sub> )[·16]  | this board                                    |

An embedding is one row of fourteen. Only fact, raster, cube and tree ids hash the observed bytes; bundle, raster, state and track ids hash lists of other ids; an entity id hashes a name. Grammars and hash rules as coded at emem 18adb67 (research/repro/v10/algorithms.mnd).

`pixel: col = [i + (E - X)/sx], row = [j + (Y - N)/sy] in UTM 43N · p = (DN + o)·10−4, o = −1000 if baseline ≥ 04.00 (PC) · NDVI = (ps − pa)/(ps + pa) · SCL ∈ {0,1,8,9,10} → next scene`



By our reconstruction of its read path, emem range-reads **0.058 %** of the scene to sign one NDVI value; three 64 KiB COG headers and one tile each of B04, B08 and SCL. It stores the 1,115-byte record, not the tiles. The next agent receives 84 characters (46 GPT-4 tokenizer tokens).

We hashed the Earth Search (AWS) copies of the B08 and B04 COGs into two signed pointer notes [\\$2](#) [\\$3](#): all 292 chunks of each (header + 291 tiles; 238.7 MB, 232.9 MB) under one Merkle root; the files stay on AWS. The tile holding the cell is emem:tree:kh1qtqrdbb6jponqn4gv72if7e#row=108 (row = leaf index); its 1,586,895 bytes hash to that row. These are not the Planetary Computer file the NDVI was read from (281.9 MB, offset not pre-subtracted), and emem records no hash of that file.

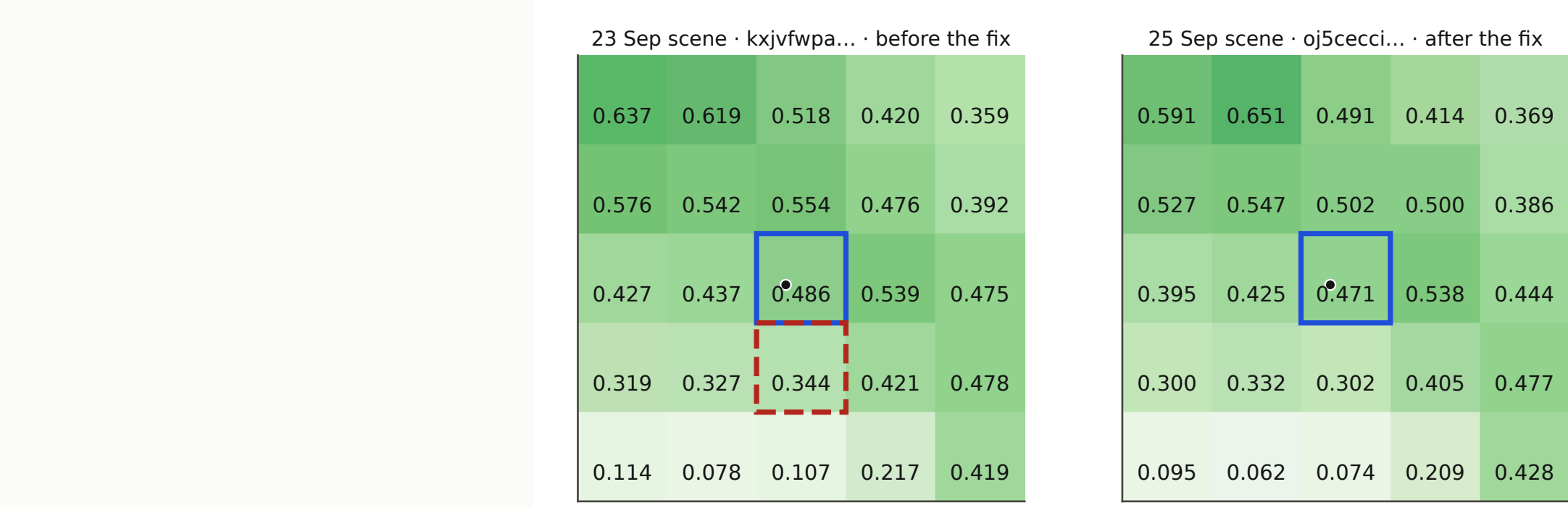
```
$4 emem:fact:defi.zb572.xoso.zb1ec:ojs5ceccile62vm6hedpuk67cjt2pqcz73mtsuyffgakxbmyaa
band indices.ndvi      day 20721 since 1970-01-01 = 25 Sep 2026
scene S2A_MSIL2A_20260925T054251_R00S_T43SFS · scene class 4, vegetation
DN   B08 3502 B04 1900 B0A offset −1000 (PC, baseline ≥ 04.00)
value 0.4708994708994709 (B08-B04)/(B08+B04) after offset: bit-identical
signer 777ex3yi... signed_at 2026-09-28T09:06:56Z · 1,115 B · BLAKE3 → ojs5cecci...
```

Earth Search serves the same tile with the offset already subtracted (B08 2502, B04 900); NDVI is identical. Sizes: Planetary Computer asset sizes (HTTP HEAD) and range reads, 30 Sep; the pointer notes and the 1,586,895-byte tile are Earth Search copies.

### 4 · WHAT THE TRACE FOUND

#### Before 28 Sep, 162 of 200 sampled records carry a neighbouring pixel

`pre-fix row = round(j + (Y - N)/sy): the south pixel whenever the fraction ≥ ½ · fixed: [·] · prevalence 162/200, Wilson score interval, z = 1.96`



Re-reading the COGs named inside the signed records showed that the 23 Sep record [\\$7](#) carries the DNs of the pixel **10 m south** (dashed red: 2993/1972, SCL 5 bare soil, NDVI **0.3444**). The pixel that contains the point (blue) reads 0.4860. emem had rounded the fractional pixel index; GDAL takes the floor. The fix shipped on 28 Sep [\\$8](#).

#### AUDIT OF PRE-FIX SENTINEL-2 RECORDS

random sample, one per cell, of pre-fix S2 records cited on emem.dev’s agent channel: 200 cells, 164 scenes, 19 bands (192 Element84, 8 Planetary Computer)

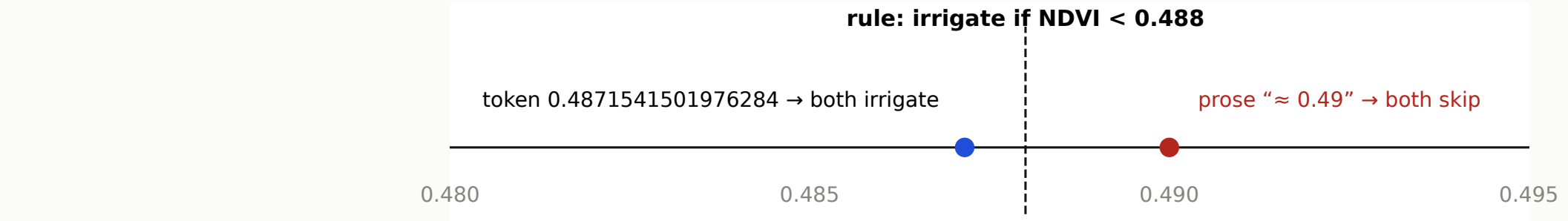
|                                                                |                             |
|----------------------------------------------------------------|-----------------------------|
| DNs match the rounded (neighbour) pixel, not the floor pixel   |                             |
| rounding and floor give the same pixel                         | <b>162 · 81 % [75–86 %]</b> |
| DNs match the floor pixel where the two rules differ           | <b>38</b>                   |
| ΔNDVI in emem’s signed change record at this cell, 23 → 25 Sep | <b>0</b>                    |
| ΔNDVI at the containing pixel                                  | <b>+0.127</b>               |
|                                                                | <b>−0.015</b>               |

Re-reading the pixel found the error; the signatures date it. No signed byte was rewritten: old tokens still resolve to exactly what was signed, so the error is reproducible and datable. Post-fix records (after 28 Sep) read the containing pixel. Wilson 95 % interval. Figure: ● query point, blue = pixel containing it, dashed red = pixel emem read. The offset is a reproducibility error, not a claim about geolocation accuracy.

### 6 · COMPACTION: WHY A PARAPHRASE IS NOT A CITATION

#### After a shared summary, 0 of 72 answers were right, yet 3 of 36 pairs agreed

`H1: P(pair agrees) > P(answer correct) · Fisher exact one-sided on 3/36 vs 0/72 → p = 0.035 · rule: irrigate = NDVI < 0.488`



|                                                                                                                                                                                                                    | CORRECT      | AGREE        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|--------------|
| full context (control)                                                                                                                                                                                             |              |              |
| shared summary, compaction under pressure                                                                                                                                                                          | <b>72/72</b> | <b>36/36</b> |
| shared summary, no pressure                                                                                                                                                                                        | <b>0/72</b>  | <b>3/36</b>  |
| wrong-but-agreeing pairs (3/36) exceed correct answers (0/72): Fisher one-sided p = 0.035 <a href="#">\$17</a> ; handoff n = 20 per arm: prose 2, dense retrieval 8, BM25 20, token bundle 20 <a href="#">\$18</a> | <b>20/72</b> | <b>15/36</b> |

The trap value 0.4871541501976284 [\\$21](#) is the value observed 17 Jul and signed 18 Jul. It is a pre-fix read of the pixel south (section 4); it tests fidelity to the cited value, not the ground. BM25 ties the bundle on short handoffs; no-pressure arm p = 0.109 (n.s.); six scoring bugs were corrected across two scorers. Related: Perez et al., ICLR 2025; arXiv 2608.11242; arXiv 2607.25066.

### 8 · RAW BANDS, THE AGENT’S OWN METHOD

#### Asked for 23 Sep raw bands, emem’s raster tool served the 25 Sep scene

`scene(t) = newest S2 item with cloud < c in [t − D, min(t + D, now)], (c, D) ∈ (40 %, 30 d), (60 %, 60 d), (80 %, 90 d) · t = 23 Sep − window 24 Aug − 30 Sep → 25 Sep (10.8 %); the 23 Sep scene (20 %) cannot win`

#### 10 PRE-REGISTERED RUNS, CLAUDE SONNET 5.5, EMEM AS THE ONLY MCP SERVER

|                                                                                                                                           |                    |
|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| chose their own bands and formula: B04, B08, (B08-B04)/(B08+B04)                                                                          | <b>10</b>          |
| noticed that both dates had returned one scene                                                                                            | <b>9</b>           |
| got 23 Sep values (via recall and backfill): ΔNDVI 0.486 → 0.471 = −0.015, “no material change” <a href="#">\$27</a> <a href="#">\$28</a> | <b>1</b>           |
| wrote “no material change” as a placeholder, saying it was not measured                                                                   | <b>5</b>           |
| answered “undetermined”, outside the allowed answers                                                                                      | <b>4</b>           |
| said “greener”, the answer the pre-fix record gives (+0.127)                                                                              | <b>0</b>           |
| wrongly said no 23 Sep scene exists                                                                                                       | <b>3</b>           |
| cited ids that re-hash · data values traced to tool results                                                                               | <b>30/30 · all</b> |

The failure was emem’s: emem\_band\_raster returned the byte-identical 25 Sep raster with no warning. The scene named inside the record is how nine agents saw it. No run used SCL.

Pre-registration blake3 30d8a1a1... hashed 13:49Z, 30 Sep, before trial 1; write tools denied. The primary test (does the pre-fix record mislead?) could not run; emem’s read tools signed new records mid-run, so trials 9–10 were not independent. research/repro/data/v9/rawband · [\\$22](#)

### 10 · WHERE EMEM LOST

#### The authors’ own scorecard

|                                                                                                                                             |                           |
|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| exact = answer = signed value · confidently wrong = answered A answer ≠ signed value · BM25 and dense retrieval top-5 over the same corpora |                           |
| CLAIM, AS SCORED BY THE AUTHORS <a href="#">\$19</a>                                                                                        | VERDICT                   |
| survives compaction where a paraphrase does not                                                                                             | <b>supported</b>          |
| beats plain context when the value fits                                                                                                     | <b>refuted</b>            |
| beats BM25 on these corpora (BM25 16/16)                                                                                                    | <b>refuted</b>            |
| single tokens save context (they cost 9.5× the LLM tokens)                                                                                  | <b>refuted</b>            |
| model agreement is evidence of correctness                                                                                                  | <b>refuted, p = 0.035</b> |
| a bundle saves context (38 chars for up to 256 facts)                                                                                       | <b>supported</b>          |
| beats peer memory products                                                                                                                  | not tested                |

Dereferenced citations: **99.2 %** exact, 0 confidently wrong (84.4 % before four fixes); dense retrieval 4/142, up to 138 confidently wrong by a median 252 m. On a retrieval miss one model abstained 74/96 and the other gave a real neighbouring-cell value 93/96 [\\$25](#).

Scope: 5 sites, 2 open 7–12B models, one host, no independent replication (labelled SAMPLE). The record lists nineteen withdrawn claims [\\$](#)